

Laura Gao: Strategy Brief

Knox Grammar · Wharton Global High School Investment Competition 2026–27 · One document to read (about 15 min) · 25 Sep 2026

The answer: We recommend "Lock, then grow."

Laura's **first cheque (\$300k, Jan 2027)** buys a ladder of US Treasury bonds, one maturing just before each \$50k payment in 2033–2042. That makes the operating commitment certain from day one; at today's ~5.15% yields it costs $\approx$ **\$298k**.

Her **second cheque (\$150k, Jan 2028)**, plus anything left over, goes into a diversified growth portfolio that becomes her **facility contribution**. The projection is $\approx$ **\$167k** (range \$118k–\$230k).

In 2031 she locks in most of that surplus so she can give co-sponsors a promise she knows she can keep: "**at least $\approx$ \$113k, likely \$153k–\$174k.**"

1. Laura's problem in three lines

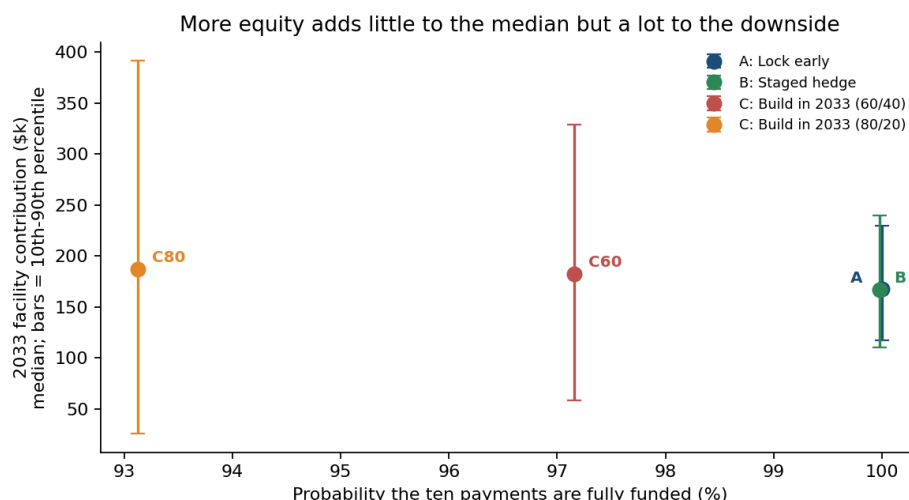
- **Money in:** \$300k (start of 2027) and \$150k (start of 2028). Nothing else in or out before 2033.
- **A promise she can't break:** ten fixed \$50k payments (2033–2042), funded by the portfolio alone "with a high degree of certainty." No co-sponsors or grants can help.
- **A hope:** contribute as much as responsibly possible to the Taiwan residency building in 2033, tell co-sponsors a credible range in 2031, and keep some flexibility.

The key insight is that these are **two different kinds of money**. Professionals treat a fixed promise like a pension fund does: secure the promise first, then invest the surplus for growth (Russell Investments 2026; Western Asset on cash-flow matching; Chhabra's "safety / market / aspirational" buckets).

2. The choice we made, and why

We compared three strategies in a model that simulates 20,000 possible market paths (interest rates and fat-tailed stock returns) and also replays every real 6-year stretch of US market history since 1928.

Strategy (base case)	Chance all 10 payments funded	Facility: bad (p10)	typical (p50)	good (p90)	History: windows funded
A: Lock early (our choice)	100%	\$118k	\$167k	\$230k	93 / 93
B: Staged hedge (50% → 100%)	99.98%	\$110k	\$167k	\$240k	93 / 93
C60: 60/40, buy bonds 2033	97.2%	\$58k	\$182k	\$329k	92 / 93
C80: 80/20, buy bonds 2033	93.1%	\$26k	\$187k	\$392k	90 / 93



Taking stock-market risk with the promised money adds only ~\$20k to the typical outcome, but creates a 3–7% chance (up to 13% in a bear market) of breaking a promise Laura cannot renegotiate.

Three reasons A wins

- 1. Bonds are unusually cheap to lock now.** With Treasuries near 5%, the bonds earn almost what a balanced portfolio is expected to earn. J.P. Morgan's 2026 long-term forecast for a 60/40 portfolio is 6.4%; Vanguard expects 4.2–6.2% a year for US stocks. Certainty costs very little.
- 2. It survives everything we throw at it:** bear markets, inflation shocks, the 1929 crash, 1970s stagflation, the 2008 crisis landing the year before purchase, and rates starting as low as 4%.
- 3. It doesn't depend on the second cheque.** Laura's income is irregular (advances, royalties, speaking). If the 2028 \$150k never arrived, A would still fund 100% of the payments; B would drop to 56% and C60 to 60%.

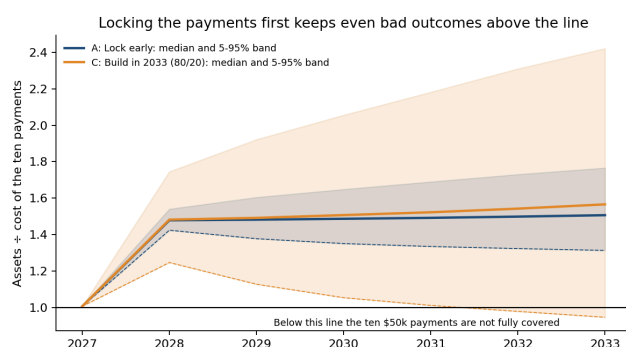


Chart 1. The worst 5% of outcomes for "build later" (orange dashed) sink below the line; "lock early" never does.

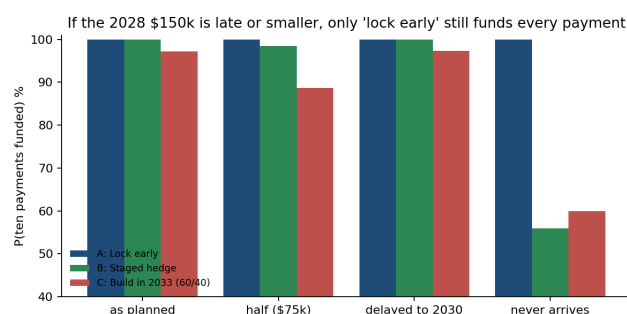


Chart 3. If the 2028 money is late, smaller, or never arrives.

Our fallback rule (this shows the choice is a decision, not a default): if Treasury yields are below ~4% in January 2027, the ladder would cost more than \$333k. In that case we switch to the staged hedge (B).

3. Our answers to the four questions the case asks

1 Operating reserve: size, composition, how it changes

- Composition:** a ladder of **regular (nominal) US Treasuries** held to maturity, one per payment. For 2033–37 use iShares iBonds term Treasury ETFs (IBTM, IBTO, IBTP, IBTQ, IBTR), which mature in December and fund the following January. For 2038–42 use individual Treasury notes/bonds.
- Size:** ~\$298k to buy in 2027. Its value in 2033 is typically ~\$413k, which is exactly the cost of the remaining payments, because the bonds and the promise move together.

- **How it changes 2033–42:** it doesn't need managing. One bond matures each year and pays \$50k. No selling, no rebalancing.
- **Not TIPS:** inflation-protected bonds don't match a promise fixed in dollars, so they would *add* risk to it.

2 What "a high degree of certainty" means

- **Mechanism:** every payment is matched by a US government bond maturing just before it is due, so markets can no longer affect it.
- **Standard:** at least 99.5% funded in every scenario we model (the "1-in-200-year" standard insurers use under Solvency II), in 93/93 historical windows, and if the 2028 money is missing. A passes all three.
- **Why not the usual 80–90%:** that benchmark suits retirees who can cut spending later (Kitces). Laura cannot adjust, so she needs an institutional standard.
- **Remaining risks, stated honestly:** US government default, a small timing gap (bonds mature Nov/Dec for January payments), and the purchase price on the actual day.

3 Facility contribution and flexibility

- **Rule:** contribute **80%** of what remains after the reserve in 2033; Laura keeps **20%** as a flexibility buffer ($\approx$ \$42k typical).
- **Projection:** \$118k / **\$167k** / \$230k (bad / typical / good), or $\approx$ \$144k in today's money.
- **Why 20%, not an arbitrary number:** (a) the USD–Taiwan dollar rate has swung -11% to $+26\%$ over 6-year periods since 2000; (b) architects carry $\sim 20\%$ cost contingency before a building is designed; (c) the case warns against committing everything.
- The buffer isn't lost. It is first in line if the residency needs more.

4 The 2031 message to co-sponsors

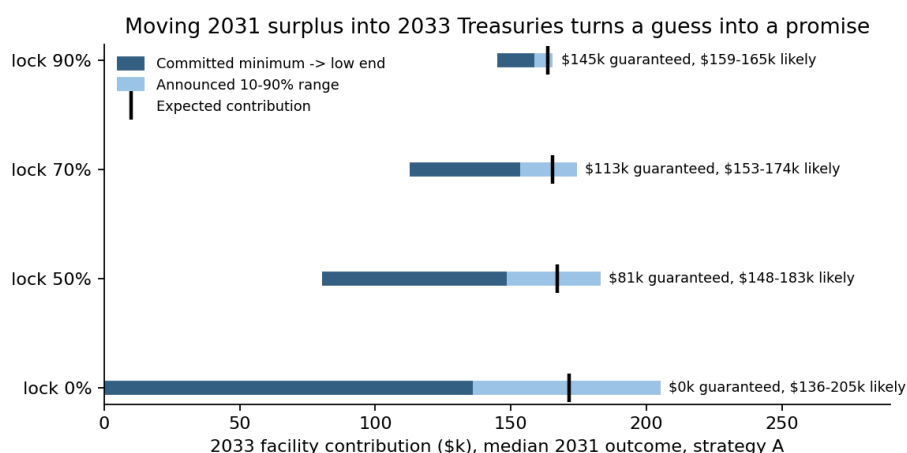


Chart 4. On the median path, moving 70% of the 2031 surplus into bonds maturing Dec 2032 turns a wide guess (\$136k–\$205k, no floor) into a firm promise: at least \$113k, likely \$153k–\$174k.

- **Rule:** at the start of 2031, move 70% of the surplus into Treasuries maturing Dec 2032. Announce "**at least \$X committed; likely \$L–\$H (about 80% confidence).**"
- **Tested:** the committed minimum was never missed in any simulated path. The low end holds 84–91% of the time, even in bear markets and historical replays.
- **Cost:** about \$6k less on average than leaving it all invested. That is the price of credibility.
- **Why it matters:** in a university fundraising field experiment, raising the visible lead gift from 10% to 67% of the goal produced a *nearly sixfold* increase in giving (List & Lucking-Reiley, 2002). A firm floor Laura can prove is worth more than a big number she might walk back.

4. The growth portfolio and WlnS

- **Mix:** 80% stocks / 20% intermediate Treasuries. The stocks are about 60% a broad global index core and 40% in 8–12 researched companies (max ~5% each).
- **Values without theme bets:** we look for quality companies connected to education, creativity, community and Asia-Pacific. We avoid thematic funds (Morningstar: only 14–18% survive and beat the market long-term) and avoid a big bet on publishing or media, which would double Laura's career risk.
- **WInS, two versions:** if ETFs are allowed, mirror the whole plan (~65% Treasury ETFs, ~35% growth). If only stocks are allowed, WInS shows the stock part of the growth portfolio, and the bond ladder is tracked in a "shadow" spreadsheet. Every Trading Note says which part of the strategy the trade serves.

5. How judges could attack this, and our reply

Attack	Reply
"It's just buying bonds."	It's a <i>rule</i> : lock when yields are $\geq \sim 4\%$, stage the hedge when below. Show the trade-off chart and tell the "two cheques" story.
"The case says she sets the reserve aside in 2033."	She <i>designates</i> it in 2033, as the case says. We buy the same bonds earlier because they're cheaper now, which is standard pension practice.
"Your model could be wrong."	The <i>payments</i> rest on bond contracts, not forecasts. Only the facility numbers depend on assumptions, and we show bad / typical / good cases for those.
"The facility number is small."	Laura can choose more risk. We show exactly what each extra \$10k of expected contribution costs in risk to her promise.

6. What the team does next

1. **Now:** check the WInS starting cash and whether ETFs are allowed (SMAApply trading document). Confirm the iBonds tickers and years on ishares.com.
2. **By 23 Oct: Trading Notes Analysis.** Plan and template: `deliverables/trading_notes_plan.md`.
3. **Team meeting:** accept, change or reject each decision (D1–D8), assign one owner per decision, and approve Laura's details in `config/client_mandate.yaml`.
4. **By 6 Nov: IPS.** Outline: `deliverables/ips_outline.md`. **Students write every sentence** (Wharton AI rule).
5. **By 4 Dec: Final Report.** Instructions arrive 9 Nov; draft structure in `deliverables/final_report_outline.md`.

7. Key assumptions (all sourced)

Assumption	Value	Source
Starting Treasury yield (6–15 yr)	5.15%	US Treasury yield curve, 23 Sep 2026
Long-run yield	4.2%	FOMC longer-run 3.2% + ~1% term premium
Yield volatility	0.9 pp / year	FRED 10-yr yield history
Stock return (base)	~6.2% / yr, 16% volatility	J.P. Morgan 2026 LTCMA; Vanguard (bear and bull cases also run)
US inflation	2.5%	10-yr breakeven (5.11% – 2.63% TIPS)
Taiwan facility-cost inflation	2.5%	Taiwan CPI 1.8–2.2% (CBC, DGBAS), plus a construction premium
History	1928–2025	Damodaran (S&P 500, 10-yr Treasury)

Everything else is in the Wharton folder if you want the detail: **decisions/** (one memo per decision), **redteam/**, **model/** (run `python -m model.laura.run`), **outputs/laura/tables.md** (all numbers). AI-assisted analysis; not submission text.